Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | ***Subject: Power Plant Engineering*** | ***Time: 03 Hours***
Code: 102610 | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Which of the following power plants produces the least amount of pollution?
(i) Coal-fired thermal plant
(ii) Nuclear power plant
(iii) Diesel power plant
(iv) Hydroelectric power plant

(b) The draught which a chimney produces is called
(i) Induced draught
(ii) Forced draught
(iii) Natural draught
(iv) Balanced draught

(c) Rankine cycle comprises of
(i) two isentropic processes and two constant volume processes
(ii) two isothermal processes and two constant pressure processes
(iii) two isentropic processes and two constant pressure processes
(iv) none of the mentioned

(d) What is the primary purpose of the boiler in a steam power plant?
(i) To convert mechanical energy into electrical energy
(ii) To heat water and convert it into steam
(iii) To cool the exhaust steam
(iv) To pump water into the condenser

(e) In a combined cycle power plant, gas and steam turbines are used together mainly to
(i) Increase the capital cost
(ii) Increase overall thermal efficiency
(iii) Improve cooling efficiency
(iv) Reduce maintenance cost

(f) Gas turbine works on
(i) Diesel cycle
(ii) Otto cycle
(iii) Dual cycle
(iv) Brayton cycle

(g) In a thermal power plant feed water heater is used to pre heat the feed water by
(i) Hot gases from boiler furnace
(ii) Steam from boiler
(iii) Hot air from air pre heater
(iv) Turbine exhaust steam.

(h) A moderator, in nuclear power plants, is a medium introduced into the fuel mass in order to
(i) Control the reaction
(ii) Reduce the temperature
(iii) Extract heat from nuclear reaction
(iv) Slow down the speed of fast moving neutrons.

(i) What is a major economic disadvantage of nuclear power plants?
(i) High fuel cost
(ii) Short plant life
(iii) High initial capital cost and long payback period
(iv) High maintenance frequency

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(j) The power tariff that varies according to the time of the day is known as
(i) Flat rate tariff (iii) Block rate tariff
(ii) Time-of-day tariff (iv) Two-part tariff

Q.2 (a) Explain in detail the classification of power plants based on different sources of energy. Provide suitable examples for each type. [7]

(b) Explain the working principle of super critical boilers and FBC with a neat sketch. [7]

Q.3 (a) What do you understand by the cogeneration of power and process heat? Explain its thermodynamic advantage. [7]

(b) A steam turbine working on Rankine cycle is supplied with dry saturated steam at 25 bar and the exhaust takes place at 0.2 bar. For a steam flow rate of 10 kg/s, determine: (i) quality steam at end of expansion, (ii) turbine shaft work, (iii) pump work and (iv) heat input. [7]

Q.4 (a) Explain integrated gasification combined cycle (IGCC) with a neat sketch. [7]

(b) A Brayton cycle (air standard) has a pressure ratio of 4 and inlet conditions of one standard atmospheric pressure and 27°C. Find the air flow rate required for 100 kW power output if the maximum temperature in the cycle is 1000°C. Assume γ = 1.4 and $C_p = 1.0 \frac{kJ}{kg-K}$. [7]

Q.5 (a) Write the major components of gas turbine power plant. [7]

(b) Explain the different types of fuel cells and their typical applications. [7]

Q.6 (a) What is meant by uranium enrichened? Describe some methods of uranium enrichment. [7]

(b) Discuss the working principle and application MHD converter with neat sketch. [7]

Q.7 (a) With neat sketch explain hydro-electric power plant. [7]

(b) Explain the various methods used to calculate the depreciation cost of the power system. [7]

Q.8 (a) Explain environment pollution due to industrial emissions. [7]

(b) Discuss about the power tariffs in the power plant. [7]

Q.9 (a) Explain the working of pressurized water reactor [7]

(b) Write short notes on ***any two*** of the following: [3.5 x 2=7]
(i) Load factor
(ii) CANDU reactor
(iii) Biogas
(iv) Ash handling system.

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